# Product Features Brief - CBWTF Waste System (CBWTF Waste Management)

This document provides a complete technical and functional overview of the CBWTF Waste System system. It is designed to assist in rebranding, selling, and deploying the software for other Common Bio-Medical Waste Treatment Facilities (CBWTF).

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## System Overview

CBWTF Waste System is a **compliance-first, end-to-end biomedical waste tracking system** built on **React, Vite, and Supabase (PostgreSQL)**. It manages the complete chain of custody for hazardous biomedical waste from generation at Health Care Facilities (HCFs) to transport, gate check-in, reconciliation, batching, treatment, and certificate generation.

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## Architecture & Technology Stack

- **Frontend Framework:** React 18 with Vite (fast, modern bundler)
- **Styling System:** Custom premium Vanilla CSS (vibrant color tokens, dark-mode first design, glassmorphism, responsive grids, micro-animations)
- **Backend-as-a-Service:** Supabase (Auth, Row-Level Security, Realtime, PostgreSQL Database)
- **Scanning Engine:** `html5-qrcode` library for real-time camera scanning of barcoded/QR-coded waste bags on mobile/desktop browsers
- **Hardware Integration:** Web Bluetooth API (`navigator.bluetooth`) to connect directly to physical weighing scales (supports standard GATT scales & custom UART/Serial Bluetooth modules)
- **Favicon & Brand Assets:** Customizable vector SVGs for crisp, lightweight scaling across devices

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## Core Modules & Functional Features

### 1. Health Care Facility (HCF) Portal

Provides hospitals and clinics with self-service tools for compliance and dispatch.

- **Dashboard:** Shows stats on waste bags generated, pending collection, and already dispatched.
- **Self-Scanning Dispatch Protocol:**
- Restricted self-scanning triggers automatically for bedded facilities (e.g. >30 beds) to ensure bags are scanned and weighed by hospital staff.
- Generates audit trails during dispatch to driver routes.
- **Facility Certificates Section:** Allows HCFs to view, print, or download official CBWTF Treatment

Certificates specific to their facility.

### 2. Driver mobile web app

Web-based workflow tailored for drivers picking up waste.

- **Route Management:** Drivers select their vehicle, check in with GPS geolocation coordinates, and start their daily collection trip.
- **QR Code Bag Pickup:** Scans bag labels, validates ownership, and checks status.
- **Live Bluetooth Scale Integration:** Pairs with digital weighing scales over Web Bluetooth to auto-capture exact bag weights (metric/imperial parsed to kg).
- **Offline Sync Queue:** Caches collection scans and GPS points in browser memory, auto-synchronizing to Supabase when returning to mobile coverage.
- **Manifests:** Generates collection manifests (Yellow, Red, Blue, White bag counts and weights) signed off on pickup.

### 3. Plant Operations Module

Manages gate arrivals, batching, and waste processing.

- **Gate Scan:** Verifies vehicle arrivals and scans incoming bags at the plant gate to match against driver manifests.
- **Reconciliation Engine:** Compares bag counts and weights scanned at the HCF against those scanned at the plant gate. Automatically triggers discrepancy alerts for missing bags or weight variations.
- **Batch Processing:** Groups reconciled bags into batches with unique identifiers, preparing them for final treatment.
- **Treatment & Disposal:** Mark batches as treated via compliant methods (Autoclave, Incineration, Microwave, Chemical, Hydroclave) and print/save legal disposal certificates.

### 4. Regulatory & Administrative Control Panel

High-level command center for CBWTF admins and pollution control regulators.

- **Compliance Dashboard:** Real-time metrics, charts (pie/bar) showing category-wise waste volume, and active driver maps.
- **HCF Registry:** Manage bed capacity, district configurations, facility codes (HCF0001), and hospital types (bedded/non-bedded).
- **User Management:** Create and manage profiles with strict role scopes (Plant Head, Manager, Driver, HCF Staff, Regulatory).
- **Discrepancy Resolution:** Worklist of flagged weight mismatches or missing bags with operators entering resolution comments to close cases.
- **Form IV Reports Engine:** Generates monthly and annual regulatory compliance reports, downloadable in printable formats.
- **Immutable Audit Trail:** Log of all scan events, GPS check-ins, and operator actions, secured by database row-level security.

## Database Schema (Supabase PostgreSQL)

The backend is structured into ten main relational tables:

1. **`profiles`**: User details mapped to roles (`plant_head`, `plant_manager`, `driver`, `regulatory`, `hcf`) and linked to specific hospitals.
2. **`hospitals`**: Registry of Health Care Facilities (district, beds configuration, codes, bedded status).
3. **`vehicles`**: Fleet tracking (vehicle numbers, status, assigned drivers).
4. **`routes`**: Active trip logs for drivers (trip date, driver name, vehicle).
5. **`bags`**: Individual waste bags (barcodes, category [Yellow/Red/White/Blue], weight, collection status, batch links).
6. **`scan_events`**: Immutable ledger recording scanning time, GPS, and operator ID at every step of custody.
7. **`manifests`**: Driver pick-up sheets showing count/weight breakdown per site.
8. **`batches`**: Staged groupings of bags ready for treatment.
9. **`discrepancies`**: Log of weight mismatches or missing bags.
10. **`audit_logs`**: Regulatory-grade immutable log tracking all actions.

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## Rebranding Checklist (For Selling to Another CBWTF)

To white-label this application for another company, update the following:

1. **Favicon & Logo:** Replace `/public/favicon.svg` and `/public/logo.svg` with the new company's design, and update `<Logo />` in `src/components/Logo.jsx`.
2. **Brand Text:** Update the texts "CBWTF Waste System" in `src/components/Logo.jsx`, `index.html`, and `AppLayout.jsx` with the new company name.
3. **Certificate Details:** Customize corporate details (address, facility description) in the printable HTML generation script in `src/lib/certificate.js`.
4. **Supabase Connection:** Set up a new Supabase database, run `supabase_schema.sql` in the SQL Editor to recreate tables/RLS policies, and update the environment variables (`.env`) with the new project URL and anon keys.